

Mathematics course sample

THEOREM • `math:cat:functors-naturality:th:equivalence-criterion`

Criterion for an equivalence

For small categories, a functor is an equivalence exactly when it is fully faithful and essentially surjective. The reverse implication uses the axiom of choice to select a representative and an isomorphism for each target object.

Proof

Suppose F has a quasi-inverse G and natural isomorphisms $u : \text{id} \Rightarrow GF$, $v : FG \Rightarrow \text{id}$. Essential surjectivity follows from v . The functors F and G are faithful because GF and FG are naturally isomorphic to identity functors. For an arrow $h : FA \rightarrow FB$, set $f = u_B^{-1}G(h)u_A$. Naturality of u gives $GF(f) = u_B f u_A^{-1} = G(h)$, and faithfulness of G yields $F(f) = h$. Thus F is full.

Conversely choose for each D an object GD and isomorphism $v_D : F(GD) \rightarrow D$. Full faithfulness defines $G(h)$ uniquely by $F(G(h)) = v_E^{-1} h v_D$ for $h : D \rightarrow E$. Uniqueness proves functoriality, and this equation makes v natural. Define $u_A : A \rightarrow GFA$ as the unique lift of v_{FA}^{-1} . These are isomorphisms because fully faithful functors reflect isomorphisms. Applying F and using the displayed equations verifies naturality of u ; faithfulness then gives it in $\mathcal{C}$. Hence G is a quasi-inverse.

Finite lists form a monad on Set

Let $T(X)$ be the set of finite lists of elements of X , including the empty list. Define $T(f)$ by applying f to every entry, $\eta_X(x) = [x]$, and μ_X by concatenating a finite list of finite lists. Verify the monad laws.

Solution

Applying functions entrywise respects identities and composition, so T is a functor. Inserting singleton lists and concatenating lists both commute with entrywise application of any function, proving naturality of η and μ .

For a list $[x_1, \dots, x_n]$, $T\eta$ yields $[[x_1], \dots, [x_n]]$, whose concatenation is the original list. The other unit map makes the one-element outer list $[[x_1, \dots, x_n]]$, with the same concatenation. For a list of lists, either first concatenating the inner lists within each middle list or first concatenating the middle lists produces the same sequence of entries in the same order. This proves associativity, including empty inner and outer lists.

Quadratic reciprocity

For distinct odd primes p, q ,

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{((p-1)/2)((q-1)/2)}.$$

Proof

Put $m = (p-1)/2$ and $n = (q-1)/2$. First, for any positive odd a coprime to p , write $ai = p[ai/p] + r_i$ for $1 \leq i \leq m$. Let $s_i = \min(r_i, p - r_i)$. Let N count the residues $r_i > p/2$. As in @math:num:quadratic-residues:th:gauss-lemma, the s_i permute $1, \dots, m$. Reducing the sum of the displayed equations modulo 2, using that a, p are odd and that $\sum r_i \equiv \sum s_i + N \pmod{2}$, gives $\sum_{i=1}^m [ai/p] \equiv N \pmod{2}$. Hence the symbol has sign $(-1)^{\sum [ai/p]}$.

Now count lattice pairs $1 \leq i \leq m, 1 \leq j \leq n$. None satisfy $qi = pj$, since p, q are distinct primes and $i < p, j < q$. Exactly $[qi/p]$ pairs for a fixed i satisfy $pj < qi$; this floor is at most n . Exactly $[pj/q]$ pairs for fixed j satisfy $qi < pj$. Therefore the two floor sums add to mn . The sign formula for (q/p) and (p/q) proves reciprocity.

Construct infinitely many Pell solutions

For $x^2 - 2y^2 = 1$, start with $(x_0, y_0) = (1, 0)$ and define $x_{n+1} = 3x_n + 4y_n, y_{n+1} = 2x_n + 3y_n$. Prove this gives infinitely many distinct nonnegative integer solutions, and compute the first three with $y > 0$.

Solution

The recurrence multiplies $x_n + y_n\sqrt{2}$ by $3 + 2\sqrt{2}$, whose norm $(3 + 2\sqrt{2})(3 - 2\sqrt{2})$ is 1. Equivalently expansion gives $(3x + 4y)^2 - 2(2x + 3y)^2 = x^2 - 2y^2$. Thus every iterate is an integer solution. We obtain $(3, 2)$, $(17, 12)$, and $(99, 70)$. For $n \geq 0, x_n \geq 1, y_n \geq 0$, so $y_{n+1} > y_n$, proving distinctness. This construction does not assert that every Pell solution has been classified.

A recursive approximation to a square root

Prerequisites: `@math:uganalysis:sequences:th:monotone-convergence`,
`@math:uganalysis:sequences:th:limit-laws`.

Fix $a > 0$ and $x_1 > 0$, and set $x_{n+1} = \frac{1}{2}(x_n + a/x_n)$. Prove that $x_n \rightarrow \sqrt{a}$.

Solution

All terms are positive. Squaring the recursion gives

$$x_{n+1}^2 - a = \frac{1}{4}(x_n - a/x_n)^2 \geq 0.$$

Thus $x_n^2 \geq a$ for $n \geq 2$, and $x_{n+1} - x_n = (a - x_n^2)/(2x_n) \leq 0$ for those indices. The tail is decreasing and nonnegative, so it has a limit $L \geq 0$. In fact $x_n \leq x_2$ and $x_n^2 \geq a$ imply $x_n \geq a/x_n \geq a/x_2 > 0$, hence $L > 0$. Passing to the recursion gives $L = (L + a/L)/2$, so $L^2 = a$. The positive solution of this equation is unique because positive squaring is strictly increasing; it is $\sqrt{a}$. This argument also proves existence of the positive square root without presupposing it.

Algebra, order, and interval additivity of the Riemann integral

Prerequisites: @math:uganalysis:riemann-integration:th:darboux-criterion.

For Riemann integrable $f, g : [a, b] \rightarrow \mathbb{R}$ and real α, β , the function $\alpha f + \beta g$ is integrable and

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g.$$

If $f \leq g$, then $\int_a^b f \leq \int_a^b g$. For $a \leq c \leq b$, the restrictions are integrable and

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Also $|f|$ is integrable and $|\int_a^b f| \leq \int_a^b |f| \leq (b - a) \sup_{[a,b]} |f|$.

Proof

The oscillation of $f + g$ on an interval is at most the sum of their oscillations. Choose partitions with small Darboux gaps for f and g and take a common refinement; this proves sum integrability. The inequalities

$$L(f, P) + L(g, P) \leq L(f + g, P) \leq U(f + g, P) \leq U(f, P) + U(g, P)$$

then squeeze the sum integral to the sum of the integrals as the two gaps tend to zero. Positive scaling scales both sums; negative scaling interchanges upper and lower sums with the appropriate sign. These facts give linearity. If $f \leq g$, every lower sum for f is at most the corresponding lower sum for g , proving integral order.

Insert c in a partition. Its upper and lower sums are sums of the corresponding quantities on the two subintervals. Refinement cannot increase the gap, and each subinterval gap is nonnegative, so integrability on the full interval implies integrability on each part. Taking partitions with gaps tending to zero yields interval additivity. Finally, $||f(x)| - |f(y)|| \leq |f(x) - f(y)|$ bounds the oscillation of $|f|$ by that of f , giving its integrability. Apply order to $-|f| \leq f \leq |f|$ and to $|f| \leq \sup |f|$ to obtain the bounds.

Local existence and uniqueness by successive approximation

Suppose f is continuous on $[t_0 - a, t_0 + a] \times \overline{B}(y_0, b)$ with $a, b > 0$, satisfies $\|f\| \leq M$, and is Lipschitz in y there with constant L . Choose $h > 0$ with $h \leq a$, $hM \leq b$, and $hL < 1$. There is a unique solution on $[t_0 - h, t_0 + h]$ among continuous curves staying in that ball, with $y(t_0) = y_0$.

Proof

Let X be the continuous curves on this interval satisfying $\|y(t) - y_0\| \leq b$, with uniform distance. Uniform Cauchy sequences have pointwise limits in complete $\mathbb{R}^n$, converge uniformly to them, and uniform limits are continuous; the bound is closed. Thus X is complete. Define $(Ty)(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds$. The bound $hM \leq b$ makes T map X into itself, and $\|Tu - Tv\|_\infty \leq hL \|u - v\|_\infty$. Starting with the constant curve, successive differences shrink geometrically; summing that bound makes the iterates Cauchy. Their limit y satisfies $Ty = y$ by the same Lipschitz estimate. The fundamental theorem of calculus gives $y' = f(t, y)$. If u, v are fixed points, their distance is at most hL times itself, so they coincide. Every solution in the ball satisfies the integral equation.

Construct a convergent series solution

Solve $y'' + xy = 0$ with $y(0) = 1$, $y'(0) = 0$ by a power series. Give its recurrence, first four nonzero terms, and a convergence justification.

Solution

Writing $y = \sum_{n \geq 0} a_n x^n$, the constant coefficient gives $a_2 = 0$ and for $n \geq 1$,

$$a_{n+2} = -\frac{a_{n-1}}{(n+2)(n+1)}.$$

With $a_0 = 1$, $a_1 = a_2 = 0$, only multiples of 3 survive. Thus $y = 1 - x^3/6 + x^6/180 - x^9/12960 + \dots$. The ratio of consecutive nonzero terms in absolute value is $|x|^3/[(3k+3)(3k+2)]$, tending to zero for every fixed x . The series has infinite radius and may be differentiated termwise on compact intervals. The recurrence therefore verifies the equation and initial values; uniqueness of the linear initial-value problem proves it is the desired solution.

An infinite cofinite space

Let X be infinite and let the open sets be $\emptyset$ and the subsets with finite complement. Verify the topology axioms and describe the closed sets, the closure of an infinite subset, and whether the space is Hausdorff.

Solution

A nonempty union includes one cofinite set, so its complement is a subset of a finite set. A finite intersection of cofinite sets has complement a finite union of finite sets. Thus these sets form a topology. The closed subsets are X and the finite subsets. The only closed set containing an infinite subset A is X , so $\overline{A} = X$. Any two nonempty open sets intersect: their intersection has finite complement and an infinite underlying set. Distinct points therefore cannot have disjoint open neighborhoods, so the space is not Hausdorff.

Compactness in metric spaces

For a metric space X , the following are equivalent: compactness; sequential compactness (every sequence has a convergent subsequence); completeness plus total boundedness.

Proof

Compactness implies total boundedness by the cover of all radius- ε balls. It implies sequential compactness: if a sequence had no cluster point, every point would have a neighborhood containing only finitely many terms (counting indices); a finite subcover would then contain only finitely many terms in total. A cluster point supplies a subsequence using radius- $1/n$ balls. A Cauchy sequence with a convergent subsequence converges to the same limit, so compactness implies completeness.

Completeness and total boundedness imply sequential compactness: successively select infinite subsequences lying in single balls of radii 2^{-k} ; a diagonal subsequence is Cauchy and converges.

Finally assume sequential compactness. Failure of total boundedness yields an infinite sequence of mutually ε -separated points, impossible. Every open cover has a Lebesgue number: otherwise choose x_n whose radius- $1/n$ ball fits in no cover member; a convergent subsequence and a neighborhood of its limit contradict this property. A finite cover by balls smaller than that Lebesgue number therefore refines to finitely many members of the given cover. Hence X is compact.

A real improper integral from residues

Prerequisites: `@math:complex:residues-zeros:th:residue-theorem`, `@math:complex:contour-integration:th:ml-estimate`.

Evaluate $\int_{-\infty}^{\infty} dx/(1+x^4)$ by integrating over an upper semicircle, and justify passage to the limit.

Solution

The upper-half-plane poles are $z_1 = e^{i\pi/4}$ and $z_2 = e^{3i\pi/4}$. They are simple with residues $1/(4z_j^3)$, whose sum is $-i/(2\sqrt{2})$.

To justify the residue identity for the semicircular contour, take $R > 1$, excise small disjoint disks around the two poles, and join their circles to the outer boundary by nonintersecting cuts. Apply Cauchy's theorem to the boundary of the resulting simply connected region; one can first move its boundary slightly inward and then use continuity to take the limit. The two orientations along each cut cancel. The outer integral therefore equals the sum of the positively oriented small-circle integrals. On each small circle the holomorphic remainder integrates to zero and the simple principal part integrates to $2\pi i$ times its residue. Hence the outer contour integral is $\pi/\sqrt{2}$.

On the semicircular arc, $|1+z^4| \geq R^4 - 1$, so its integral has magnitude at most $\pi R/(R^4 - 1) \rightarrow 0$. The real integrand is continuous, positive, and bounded by x^{-4} for $|x| \geq 1$, so both one-sided improper integrals converge. Therefore the limiting real integral, not merely its principal value, equals $\pi/\sqrt{2}$.

Locally uniform limits remain holomorphic

Prerequisites: `@math:complex:contour-integration:th:cauchy-formula`, `@math:complex:analytic-consequences:th:taylor-cauchy`.

Suppose f_n are holomorphic on an open set U and converge uniformly on every compact subset of U to f . Then f is holomorphic. Moreover, derivatives of every fixed order converge locally uniformly to the corresponding derivatives of f .

Proof

Fix a closed disk contained in U . Uniform convergence on that disk makes f continuous there. Uniform convergence on its boundary permits passage to the limit in Cauchy's formula, giving the same integral representation for f at every interior point. Expanding the kernel geometrically on any smaller disk, as in the Taylor derivation, proves that this representation is holomorphic.

For derivatives, differentiating the kernel under the contour integral is justified uniformly on a smaller concentric disk because the distance to the contour is positive. The difference formula for order k has kernel $k!/(z - \zeta)^{k+1}$, so its magnitude is bounded by a fixed constant times the uniform boundary error $\sup |f_n - f|$. This tends to zero uniformly on the smaller disk. A compact subset of U is covered by finitely many such smaller disks, proving local uniform convergence of the derivatives.

Eisenstein's irreducibility criterion

Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $n \geq 1$. If a prime p satisfies $p \nmid a_n$, $p \mid a_i$ for $i < n$, and $p^2 \nmid a_0$, then f is irreducible in $\mathbb{Q}[x]$.

Proof

We first justify the needed form of Gauss's lemma. A primitive integer polynomial has coefficients with gcd 1. A product of primitive polynomials is primitive: otherwise a prime dividing all product coefficients would make their reductions have zero product in the domain $\mathbb{F}_p[x]$, although neither reduction is zero. Clearing denominators and dividing contents now shows that a factorization over $\mathbb{Q}$ of a primitive integer polynomial yields a factorization over $\mathbb{Z}$: the rational scalar left over must be an integer because the product is primitive, and must have absolute value 1 by primitiveness.

Divide f by its content; this does not alter the Eisenstein hypotheses because its content is prime to p . If $f = gh$ with both degrees positive, reduction modulo p is a nonzero multiple of x^n . Both leading coefficients remain nonzero modulo p , so both reduced factors have their original positive degrees. A divisor of x^n in $\mathbb{F}_p[x]$ is a scalar times a power of x ; thus both constant coefficients are divisible by p . Their product a_0 is then divisible by p^2 , a contradiction.

The intermediate fields of a biquadratic extension

Determine the Galois group and all intermediate fields of $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$.

Solution

The degree of $\mathbb{Q}(\sqrt{2})$ is 2. If $\sqrt{3} = a + b\sqrt{2}$ for rational a, b , squaring gives $3 = a^2 + 2b^2 + 2ab\sqrt{2}$, so $ab = 0$. The alternatives require a rational square to equal 3 or $3/2$, both impossible by prime factorization of numerator and denominator. Hence $[L : \mathbb{Q}] = 4$, with basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$.

The four sign choices for $\sqrt{2}, \sqrt{3}$ define distinct automorphisms, checked by multiplication in this basis. They form $C_2 \times C_2$. The extension is separable in characteristic zero. For normality, the orbit of any $a \in L$ under the four sign changes gives a polynomial $\prod_b (x - b)$, over its distinct orbit elements. Its coefficients are fixed by both sign changes and hence lie in $\mathbb{Q}$, as the displayed basis shows. The minimal polynomial of a divides this polynomial and therefore splits in L . Thus $L/\mathbb{Q}$ is normal. The three subgroups of order two have fixed fields $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, and $\mathbb{Q}(\sqrt{6})$ respectively. By the Galois correspondence the full list is these three fields together with $\mathbb{Q}$ and L .

Completeness of L^p

Prerequisites: `@math:analysis:lp-spaces:th:minkowski`, `@math:analysis:lebesgue-integration:th:fatou`, `@math:analysis:lebesgue-integration:th:monotone-integrals`.

For every measure space and $1 \leq p \leq \infty$, the normed space L^p is complete.

Proof

First let $p < \infty$ and let (f_n) be Cauchy in norm. Select a subsequence with $\|f_{n_{k+1}} - f_{n_k}\|_p \leq 2^{-k}$. Put $h_k = |f_{n_{k+1}} - f_{n_k}|$. Minkowski bounds every finite sum of the h_k by $\sum 2^{-k}$ in L^p . Monotone convergence of their p th powers gives $H = \sum h_k \in L^p$, so H is finite almost everywhere. The telescoping series therefore converges absolutely almost everywhere to a measurable f (set it to zero on the measurable divergence set). Its tail is bounded by $\sum_{j \geq k} h_j$, whose L^p norm is at most $\sum_{j \geq k} 2^{-j}$ by Fatou and the finite-sum bound. Thus $f_{n_k} \rightarrow f$ in L^p , and the Cauchy property gives convergence of the original sequence.

For $p = \infty$, take the same subsequence. Outside the union of countably many null exceptional sets, each pointwise increment is bounded by 2^{-k} . The series converges uniformly there, yielding an essentially bounded measurable limit with the same norm-tail bound. Again the original Cauchy sequence converges.

A compact operator with a spectral value that is not an eigenvalue

Prerequisites: @math:analysis:spectral-theory:df:spectrum-compact, @math:analysis:spectral-theory:df:adjoint.

On ℓ^2 , define $(Tx)_n = x_n/n$. Show that T is compact and self-adjoint and that

$$\sigma(T) = \{0\} \cup \{1/n : n \geq 1\}.$$

Determine which of these values are eigenvalues.

Solution

Truncating after coordinate N gives a finite-rank operator T_N with $\|T - T_N\| = 1/(N + 1)$. A norm limit of compact operators is compact: for a bounded sequence, repeatedly extract subsequences for each T_N , take a diagonal subsequence, and use the operator-norm approximation to show its T -images are Cauchy. Completeness supplies the image limit. The diagonal coefficients are real, so the inner-product formula shows $T = T^*$.

Each $1/n$ has eigenvector e_n . The operator is injective, so zero is not an eigenvalue, but it is not onto: $(1/n)$ lies in ℓ^2 and its only formal preimage is $(1, 1, \dots)$, which does not. Finally, if λ lies outside the displayed closed set, $\inf_n |1/n - \lambda| > 0$. Coordinatewise division by $1/n - \lambda$ defines a bounded inverse of $T - \lambda I$. Hence no other spectral values occur.

Concatenation defines a group

The operations in @math:top:homotopy:df:fundamental-group make $\pi_1(X, x_0)$ a group.

Proof

Concatenating two endpoint-preserving homotopies on the two halves of I gives a homotopy of concatenations; finite closed pasting proves continuity. Thus multiplication is well-defined. If two parameterizations traverse the same finite list of paths at different positive speeds, interpolate their nondecreasing piecewise-linear time-change functions; composition gives an endpoint-preserving homotopy. This identifies $(\alpha * \beta) * \gamma$ with $\alpha * (\beta * \gamma)$ and removes a stationary interval, proving associativity and the identity law.

For cancellation, set $L_s = 1 - s$ and define a loop by $\alpha(2tL_s)$ for $t \leq 1/2$ and $\alpha(2(1-t)L_s)$ for $t \geq 1/2$. At $s = 0$ it is $\alpha * \bar{\alpha}$; at $s = 1$ it is constant at x_0 . The same construction with the reversed path proves the other inverse law.

Homology of spheres

Compute the reduced integral homology of S^n for every $n \geq 0$.

Solution

For $n = 0$, the sphere is two points, and the augmentation $H_0(S^0) = \mathbb{Z}^2 \rightarrow \mathbb{Z}$ has kernel $\mathbb{Z}$. Positive-degree homology vanishes: each point is contractible and a singular simplex, having connected domain, lies in one component.

For $n \geq 1$, cover S^n by the complements of its north and south poles. Both sets are contractible (stereographic projection identifies each with $\mathbb{R}^n$), and their intersection deformation retracts onto the equatorial S^{n-1} . The reduced Mayer–Vietoris sequence therefore gives $\widetilde{H}_k(S^n) \cong \widetilde{H}_{k-1}(S^{n-1})$ for $k \geq 1$. The sphere S^n is path connected for $n \geq 1$, so its reduced degree-zero homology is zero. Induction yields $\widetilde{H}_k(S^n) = \mathbb{Z}$ for $k = n$ and zero for all other nonnegative k . This calculation precedes the definition of degree and does not rely on cellular boundary degrees.

Stokes theorem on manifolds

Let M be an oriented smooth n -manifold with boundary, $n \geq 1$, and let ω be a compactly supported $(n-1)$ -form smooth up to the boundary. Then

$$\int_M d\omega = \int_{\partial M} i^* \omega,$$

where i is inclusion and the boundary has the outward-first orientation.

Proof

A subordinate partition of unity writes ω as a finite sum of forms supported in individual coordinate charts. Because $\sum \rho_i = 1$, we have $d\omega = \sum d(\rho_i \omega)$. It suffices to prove the identity for one chart, extending coefficients by zero outside its compact support.

In an interior chart write $\omega = \sum_j (-1)^{j-1} a_j dx^1 \wedge \cdots \widehat{dx^j} \cdots \wedge dx^n$. Then $d\omega = (\sum_j \partial_j a_j) dx^1 \wedge \cdots \wedge dx^n$. Every integral of $\partial_j a_j$ over $\mathbb{R}^n$ is zero by iterated integration and compact support.

In a boundary chart use $x^n \geq 0$. The tangential derivative integrals still vanish, while integration in x^n gives $-\int a_n(x', 0) dx'$. Restricting ω to the boundary keeps only its $j = n$ term; the outward vector is $-\partial_n$, and the outward-first rule gives exactly this same signed integral. Sum over the charts. This uses the partition theorem and Euclidean change of variables, not an unproved cancellation of arbitrary boundaries.

Compute the de Rham cohomology of the circle directly

View S^1 as $\mathbb{R}/(2\pi\mathbb{Z})$. Compute its de Rham cohomology and give an explicit criterion for a one-form to be exact.

Solution

Every smooth one-form pulls back to $a(t)dt$ with smooth 2π -periodic a . There are no two-forms on a one-manifold, so all one-forms are closed. Exactness means $a(t) = f'(t)$ for a smooth periodic function, which forces $\int_0^{2\pi} a(t)dt = 0$. Conversely if this integral is zero, $f(t) = \int_0^t a(s)ds$ is periodic and gives a primitive on the circle. Thus integration identifies $H_{\text{dR}}^1(S^1)$ with $\mathbb{R}$; a representative generator is the angular form of integral 2π or its normalized version of integral 1. Closed zero-forms are constants because the circle is connected, so $H^0 = \mathbb{R}$. Higher groups vanish by dimension.